AA200 - Applied Aerodynamics
Winter Quarter 2025-26
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PROBLEM SET 5 **Due Date: March 15, 2026, 11:59 pm**

NOTE: This assignment contains one extra problem (Problem 4) which will not be graded but for which solutions will be provided, in case it helps with your preparation for the final exam. At most, a multiple choice question could be drawn from the material in Problem 4 and the corresponding class notes.

Problem 1. [35 points] *Load distribution of a wing with finite span.* If the load distribution on a finite-span wing is as shown (as seen from behind the wing), sketch the trailing vortex system, indicating the magnitude and direction of the trailing vortices. On the plane defined by the wing and the direction of the free stream, can you come up with a graphical representation (you can create graph/s in MATLAB or Python) of the downwash/upwash created by this vortex system? A qualitative description is all that is asked for, but you may compute (and appropriately non-dimensionalize) as it may be easier to use MATLAB/Python plotting capabilities than sketching the result by hand.

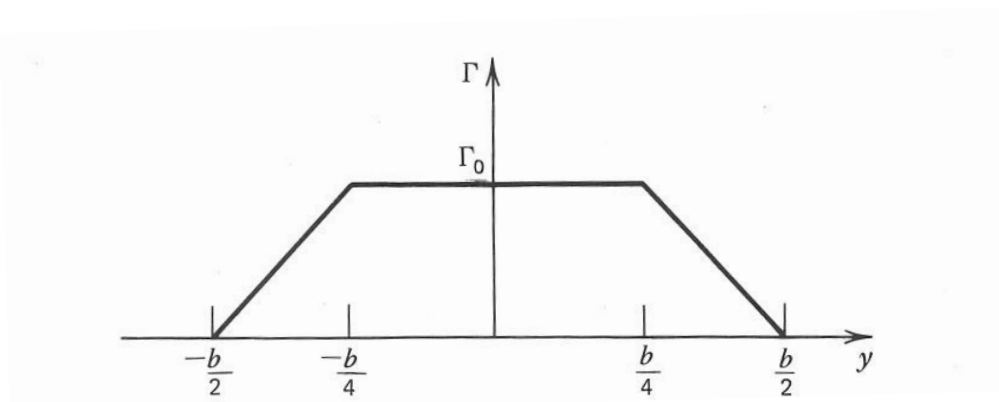

Figure 1: Spanwise circulation distribution for problem 1.

Problem 2. [35 points] For an untwisted constant-section wing of elliptical planform and aspect ratio 5, find the sectional and total lift and drag coefficients, according to lifting-line theory. Take the lift-curve slope m_0 to be 5.7, $V = 45$ m/sec, the wing loading $W/S = 1000$ N/m^2 (consider the level flight case, for which $L = W$), the altitude to be sea level, and the span $b = 10$m. Plot the absolute and induced angles of attack as functions of spanwise position, y, and calculate the power required to overcome the induced drag in this flight condition.

Problem 3. [30 points] Consider the situation where an identical wing, at identical angle of attack (as the first wing in Problem 2), is placed a few spans behind and to the right of the leading wing (in formation), such that the left wing tip of the trailing wing is at a spanwise location that coincides with the right wingtip of the leading wing. Assuming that the downwash effect of the leading wing across the entire span of the trailing wing *is a constant value equal to the induced downwash at the at the symmetry section of the trailing wing* (note: this is just an engineering approximation), answer the following questions:

- Draw a clear diagram/hand sketch (based on your answer to Problem 2) of the downwash effect of the leading wing on the trailing wing and how it changes the induced drag and the lift of the trailing wing, and
- Estimate the induced drag benefit for the trailing wing when flying in such a formation.

Problem 4.[NOT REQUIRED / NOT GRADED] *Design of a Commercial Transport Winglet for Minimum Induced Drag.* The objective of this problem set is to modify the *baseline* wing-winglet configuration for a commercial transport (provided in the MATLAB code distributed with this assignment) in order to achieve optimum performance as measured by obtaining minimum induced drag at a lift coefficient, $C_L = 0.6$. For this purpose, you will follow the approach sketched out lecture and in the paper referenced in class, which has been uploaded to canvas in the *Files->Handouts* section:

> Smith, S. C., "Trefftz-Plane Drag Minimization at Transonic Speeds", SAE General, Corporate & Regional Aviation Meeting & Exposition, SAE Paper 971478. Wichita, KS, April, 1997,

and that requires that you obtain sensitivity derivatives of the *induced drag*, D_i, of the configuration with respect to the design variables in the problem. For simplicity, we will only deal with two design variables which are the *toe-in* angles of the root and tip sections of the winglet which are highlighted in Figure 2 below. The baseline configuration is the one given to you in the example file (`ps5_analysis.m`) contained in the *AA200-VLM.zip* file and that I demonstrated in class. However, since we will be imposing a total lift constraint, we need to include a third design variable, the angle of attack of the complete configuration, α.

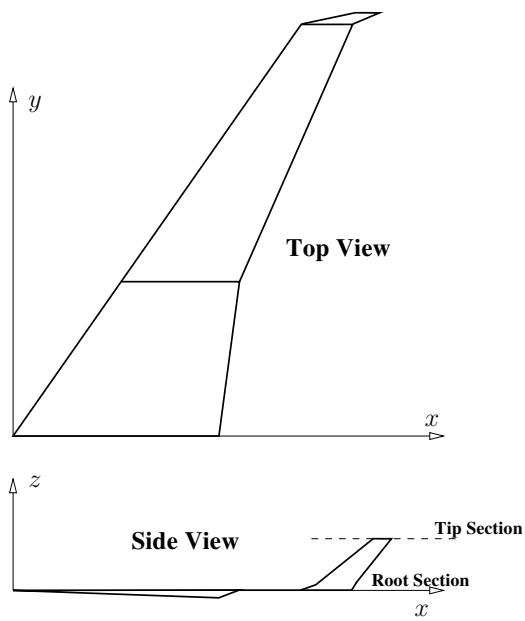

Figure 2: Top and Side View of Commercial Transport Baseline Configuration.

The aerodynamic analysis program that you will use to redesign this wing-winglet configuration is the one I showed in class.

Using the method discussed in class, you will derive the values of the optimum *toe-in* angles at the root and tip sections of the last portion of the winglet. Your solution should include a discussion of all the important steps that you followed to achieve the optimum.

Some parts of the suggested plan below are <u>required</u>, while others are optional/extra credit and are marked as such.

A suggested plan of approach includes:

1. Required. For a given wing-winglet configuration (whether it corresponds to the baseline or to a perturbed shape) you must be able to obtain the values of both the jump in potential across the wake, μ_i, and the velocity normal to the wake, ω_i, along the projected location of the wake on the Trefftz-plane. Assume that the wake travels in the direction of the free-stream from the trailing edge of the configuration. Also assume that the Trefftz-plane is far enough downstream and that the vortex filaments that intersect it are perpendicular to it, so that you can do all your downwash calculations as if the flow in the Trefftz-plane were two-dimensional. The file `ps5_geometry.m` is a sample file that you may use (or start from) to obtain the geometry of the wake in the Trefftz plane by computing the geometry of the trailing edge of the wing.

2. Required. Using the finite difference method, approximate the sensitivity derivatives required, $\frac{\partial \mu_i}{\partial x_j}$ and $\frac{\partial \omega_i}{\partial x_j}$, where x_j is the $j-th$ design variable. In this case, $j = 1, 2, 3$, corresponding to α_{root}, α_{tip}, and α. Using the finite difference method, you can obtain a reasonable approximation to these sensitivities with
$$\frac{\partial \mu_i}{\partial x_j} = \frac{\mu_{ij} - \mu_{0i}}{\delta x_j}.$$
Remember that this is only a first-order approximation so you must take a small step in the design variables δx_j so that the approximation is not inaccurate (i.e., the truncation error of the approximation dominates), yet big enough so that subtractive cancellation does not take place.

3. Required. Construct the matrix $[dic]$ and the right-hand-side vector $\{dic_0\}$ in Equations (7) and (8) of the paper by S. C. Smith.

4. The remainder of this PS is **optional** and will result in an extra credit amount of **25 points**. Invert the resulting problem (including the lift constraint) to get the optimum values of the design variables.

Overplot the spanwise load distributions of the baseline and optimum designs and answer the following questions:

1. What is the magnitude in the improvement on D_i derived from the change in these design variables?

2. In light of your calculations of ω_i and μ_i, can you explain qualitatively what is the effect of the change in the winglet twist distribution?

3. How does the optimum design off-design behavior compare with that of the baseline design? Consider the wing climb design point with $C_L = 1.0$.

If you spend enough time looking through the code provided, you will find that many of these steps are already coded in.